

Siqi Guo

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Objective: Software Development Engineer

EDUCATION BACKGROUND

- Carnegie Mellon University** **Pennsylvania, United States**
Master of Science - Electrical and Computer Engineering *Expected September 2023 - June 2025*
- Tianjin University** **Tianjin, China**
Bachelor of Science - Computer Science and Technology; GPA: 3.84/4.0; Ranking: 6/137 *September 2019 - June 2023*

SKILLS SUMMARY

- Languages:** C/C++, Python, C#, JavaScript, Markdown, Dart, SQL, Bash
- Frameworks:** NodeJS, Flask, .NET, React, Flutter, Django(Beginner), TensorFlow, PyTorch
- Tools:** GIT, Docker, MySQL, SQLServer, Qt Creator, Visual Studio Code, Xcode, Blender(Beginner)
- Platforms:** MacOS, Linux, Windows, Arduino, GCP, Kaggle

INTERN EXPERIENCE

- E2Biz SoftTech (Tianjin) Co. Ltd | Web Developer (Full-time)** *March 2023 - June 2023*
 - Product Line Supermarket System:** Utilized ASP.NET MVC to build the Warehousing Order Processing module and the Work Order Management module, designed front-end pages with Razor and KendoUI, and implemented controller to do database filtering and searching. Finally, cooperated with QA testers with several function unit tests
 - Project ESC5Scaffold:** Launched this new project. Implemented Department Planning and User & Role Management. Customized the logic of privilege setting and department transferring with the controller.
 - OurCabin System Upgrading:** Enhanced the Bug Tracking module and the FAQ section, and especially added the Excel exporting function for all forms with Aspose. Proposed and Built an Improving Item Management and Item Stage Planning subsystem with SQLServer and ASP.NET in OurCabin.

PROJECTS

- Travel App - MeiliXiu | React, NodeJS, Express, NoSQL** *Jun 2023 - In Progress*
 - Utilized Middleware and engine(Handlebars) to organize Web and templates. Introduced three-party services such as Twitter and Google Maps into the App with APIs;
 - Completed specific unit tests and integration tests with Jest.
- Flutter ToDo App | Flutter, Firebase, Cloud Firestore, Google APIs** *Jan 2023 - Mar 2023*
 - Designed UI. Implemented Firebase Authorization module and Tasks CRUD modules;
 - Performed integration testing and mocking with Flutter Driver after deploying this simple app.
- Cache Simulator with L1 and L2 level | C++, Principles of Computer Architecture** *Oct 2022 - Dec 2022*
 - Implemented a simple Cache Simulator with L1 level Cache, adjusted various cache eviction strategies such as LRU, LFU, WBWA, and WTNA, and visualized the result;
 - Implemented advanced L2 Cache with an algorithm to effectively choose a victim cache entry.
- Full-Stack Portfolio - Personal Websites | React, JavaScript, SCSS, Sanity, Node** *Mar 2022 - May 2022*
 - Customized the schemas in the backend. Used Sanity, one of the React CMSs to define content models and transform contents to frontend with APIs;
 - Developed the Web architecture with React components and Animations with Framer Motion. Used GraphQL and GROQ to fetch dynamic data from the CMS. The Portfolio was finally deployed on Netlify.
- Bilibili Database Analyst Web App | MySQL, Web Crawlers, Flask Web Development** *Sep 2021 - Nov 2021*
 - Performed web crawling on the data of Bilibili using BeautifulSoup and storage the data in the self-designed database;
 - Built a B/S web-based database application using Flask. In addition to displaying the crawled data, I also implemented CRUD functionality.
- Simulation of TCP Protocol | C++, Computer Networks, TCP Protocol** *Aug 2021 - Oct 2021*
 - Simulated and analyzed the three-way handshake protocol used by TCP with C++, conducted program refactoring, and wrote the report to reflect on the findings;
 - Verified the results of the TCP implementation by testing the packet loss rate, simulation transmission, etc.

PUBLICATIONS

- Conference Paper - Conditional generative adversarial networks: Introduction and application:** Ding, Y., & Guo, S. (November '22). In the 2nd *International Conference on Artificial Intelligence, Automation, and High-Performance Computing (AIAHPC 2022)* (Vol. 12348, pp. 258-266). SPIE.
- Conference Paper - Local-Global Defense Against Unsupervised Adversarial Attacks on Graphs:** Jin, D., Feng, B., Guo, S., et al. *Thirty-Seventh AAAI Conference on Artificial Intelligence*. In press.

HONORS AND AWARDS

- Second Prize in APCMC (Top 15%), International Level - November, 2021
- Bronze Award in National College Algorithm Design and Programming Winter Competition, National Level - March, 2021
- Third Prize in National College Mathematics Competition, National Level - November, 2020